DESformer: A Dual-Branch Encoding Strategy for Semantic Segmentation of Very-High-Resolution Remote Sensing Images Based on Feature Interaction and Multiscale Context Fusion

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***Abstract*— Global contextual information is crucial for the semantic segmentation of remote sensing (RS) images. However, the majority of current approaches depend on convolutional neural networks (CNNs). Due to the local receptive fields inherent in convolutional operations, these networks typically capture image features within limited areas and struggle to comprehend broader contextual information in the images. In this study, a dual-branch encoding approach, DESformer, is proposed, integrating transformers with CNN, to effectively capture global multiscale context information and enhance edge feature extraction. In addition, DESformer incorporates a feature interaction module (FIM) to combine local features with global representations extracted by transformers and CNN, respectively, across different resolutions. This approach enhances the capability to capture local features in RS images and improves the understanding of extensive spatial relationships. Subsequently, we employ a novel top-down approach for global supervision of the traditional feature pyramid multilevel visual integration (MVI) module, by harnessing the clear visual center information obtained from the deepest internal features. To successfully concentrate on important information and preserve sensitivity to features at various scales, the preceding shallow features are muted. In addition, FIAB-Loss, a loss function is introduced, combining a focal loss with IOU and active boundary loss (ABL). This composite loss function strengthens the model's focus on challenging-to-distinguish categories. Extensive experiments conducted on three datasets, including the semantic segmentation of lakes in the Tibetan Plateau and the ISPRS's Vaihingen benchmark, validate the efficacy of the proposed method. The experimental results indicate that the network exhibits exceptional performance in processing VHR images and accurately extracting edge features.**

***Index Terms*— Deep learning, dual-branch coding structure, remote sensing (RS), semantic segmentation.**

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I. Introduction

With the rapid development in the field of remote sensing (RS), very-high-resolution (VHR) imagery enables researchers to capture Earth's surface features with unprecedented detail and clarity, including changes in natural landscapes and the impacts of human activities [1]. Extracting valuable information from VHR images and understanding its implications becomes increasingly important in RS data analysis [2]. Against this backdrop, semantic segmentation, as an effective method for processing VHR images, has garnered extensive research and application [3]. Its main goal is to identify specific semantic categories by conducting accurate pixel-level classification within VHR images [4]. This high-accuracy pixel-level classification method renders VHR images instrumental across various domains, including urban planning, environmental monitoring, disaster evaluation, and agricultural management [5], [6], [7], [8]. The application of semantic segmentation of VHR images in multiple fields demonstrates its vast potential in VHR RS analysis.

VHR RS images have become indispensable tools in studies of land cover classification and change detection, owing to their capacity to display rich spatial details [9]. These details enhance the informational value of the images, enabling the capture of minute land features and complex terrestrial structures [10]. However, they also increase the complexity of segmentation tasks. For instance, in urban areas, VHR images can distinctly display buildings, roads, vehicles, and green spaces [11]. However, these complex spatial elements possibly bring a high degree of overlap and diversity, thus demanding more from image feature extraction [12]. In addition, processing large size and high dimensional VHR images requires powerful computational capabilities and new efficient algorithms [1]. In this context, deep learning technology has gradually become the main research tool for the semantic segmentation of VHR RS images [13]. These models can learn complex patterns and features within images, providing more precise classification results than traditional methods. However, deep learning models still face challenges in processing VHR images, such as integrating local details with global contextual information.

In the semantic segmentation task of VHR RS images, the existing methods are faced with significant challenges [14].

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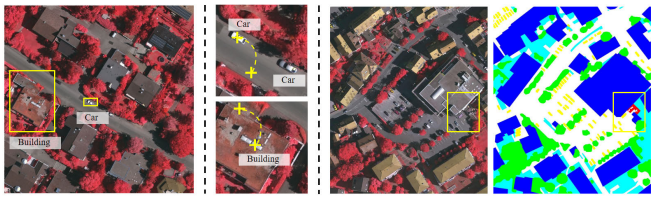


Fig. 1. Examples of challenging features in RS image segmentation. (a) Vaihingen dataset illustrates significant differences in object size, complicating the segmentation process. (b) Enlarged view highlights the necessity of contextual information for accurate segmentation. Contextual information is needed to determine whether the yellow marker is a car or part of a building. (c) Dataset exhibits an imbalance in category distribution, such as clutter (red) and buildings (blue), which further complicates the segmentation task.

These challenges primarily arise from the intricate details of VHR images and the limitations of current segmentation techniques in effectively processing this information (see Fig. 1) [15]. First, because VHR images contain rich details and complex spatial structures, the algorithm needs to be able to effectively capture local features, such as object edges, textures, and shapes [16], while also understanding the global context of the image, such as the overall layout of the scene and the spatial relationships between different objects [14]. At present, many existing semantic segmentation methods mainly rely on local feature extraction [17]. Convolutional neural networks (CNNs) are good at capturing local features due to their local receptive fields and hierarchical structures, and they have achieved significant progress in the scene classification of RS images [18]. However, they are often difficult to effectively encode the global context information of images and still face challenges in processing large-scale and complex scenes [19]. Self-attention-based transformer models are better at capturing global information than CNN because of their ability to establish long-distance dependencies across the entire image range [20]. However, the transformer model may not be as efficient as the CNN in processing local details, especially in VHR images, where it has limitations in capturing fine local features [21]. Therefore, we propose to use a dual-branch framework to extract and integrate global dependent features and local features.

Feature interaction and multiscale context fusion have been extensively studied in the field of semantic segmentation. Previous works have demonstrated the effectiveness of these techniques in enhancing segmentation accuracy. For instance, Wang et al. [22] proposed a dual encoder–decoder network for land cover segmentation, emphasizing the integration of multiscale features. Similarly, Wei et al. [23] introduced DEANet, a dual encoder with an attention network, which effectively captures the multiscale context in RS imagery. Dual-branch networks have also been explored in various forms. He et al. [24] proposed a double-branch encoder based on CNN and transformer, similar to our approach. Their work, Swin transformer embedding UNet, highlights the potential of combining CNNs and transformers for RS image segmentation. In addition, He et al. [24] developed RS 3 Mamba, a visual state space model for RS image segmentation, which further demonstrates the versatility of dual-branch architectures. In this context, our proposed DESformer model differentiates itself by introducing the multilevel visual center (MLvc) module, which uniquely integrates multiscale features from different network levels, and the FIAB-Loss function, designed to handle class imbalance and enhance boundary accuracy.

Finally, in the semantic segmentation task of VHR RS images, due to the characteristics of large image size and multiview, challenges such as boundary blurring and class imbalance are often faced [25]. VHR RS images typically cover vast geographic areas and contain objects and geomorphic features with great diversity in size, shape, and texture [26]. This huge difference in size makes it difficult to distinguish small-size objects (such as a single vehicle or a tree) from the background when pixel-level semantic segmentation is carried out, resulting in blurred boundaries. Second, class imbalance is another common problem in VHR RS image segmentation [27]. In many cases, certain categories (e.g., roads and buildings) have a much higher percentage of pixels than others (e.g., vehicles and pedestrians), causing the segmentation model to overfit frequently occurring categories while ignoring less frequent ones during training [28]. This imbalance not only impacts the model's overall segmentation performance but also decreases accuracy for small or uncommon targets [29]. Therefore, we propose using a boundary-based combined loss function to enhance the segmentation accuracy of small-scale objects and edge features.

To address these challenges, this study introduces an innovative network framework, named DESformer (see Fig. 2), which incorporates a dual-branch encoding structure combining transformers and CNN. The transformer branch of this model provides a macroscopic perspective on global contextual information, aiding in the understanding of the overall scene layout and broad spatial relationships. The main contributions of this article are as follows.

1) We introduce a feature interaction module (FIM) that enables continuous semantic fusion between the CNN and transformer branches. This approach not only enhances global awareness of local features but also improves the capture of local details within the global representations.
2) We have incorporated a centralized feature pyramid network, the multiscale visual integration (MVI) module. Based on a globally explicit centralized adjustment scheme, this structure further enhances the ability to process complex scenes within VHR images.
3) We employ a combined loss function—FIAB-Loss, which integrates global and boundary loss functions. This combined approach aims to improve the model's segmentation precision for small-scale objects and edge features while optimizing the ability to distinguish between different categories by leveraging global dependency features.

II. Related Work

Semantic segmentation is a crucial and classic task in the field of computer vision. Particularly in the semantic segmentation of VHR RS images, the integration of local features and global representations has consistently been central to the study of visual descriptors within this domain

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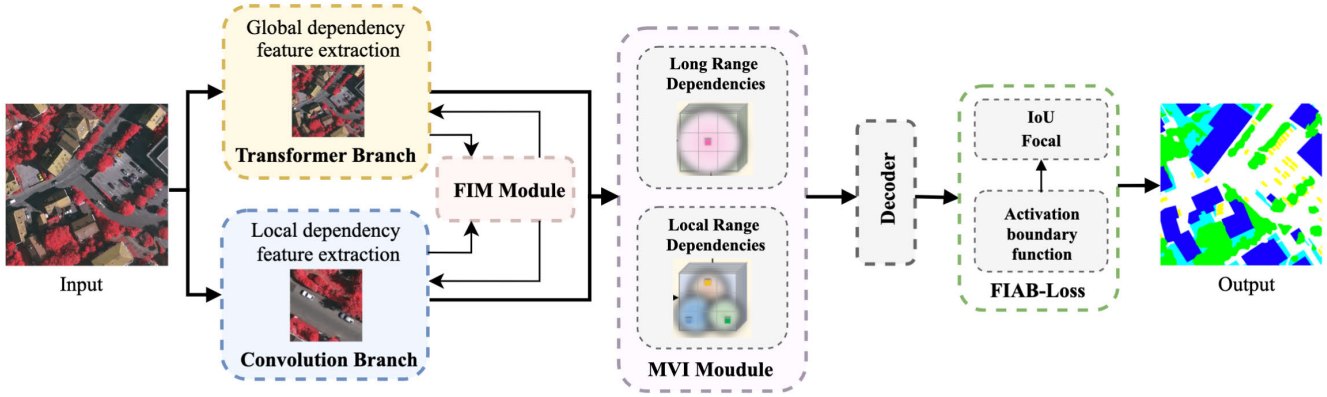


Fig. 2. Overall network architecture of our proposed DESformer includes the following components. DESformer has two branches: transformer and CNN. FIMs are embedded in each coding block of the two branches to enable continuous semantic fusion between them. Second, the MVI module based on a centralized feature pyramid network is introduced. The MVI module consists of two submodules, a local corner region (LCR) and a remote dependency, which capture local details and global dependencies, respectively. The output of two branches is fused by multiscale feature fusion. Finally, we use a composite FIAB-Loss function consisting of focus loss, IoU loss, and ABL.

[30]. Local features describe compact vector representations of the local neighborhood in an image and are fundamental to various visual algorithms [31]. Global representations, on the other hand, encompass descriptors for outlines, shape, and long-distance features of object types [32]. In recent years, one key area of focus has been how to concurrently and efficiently capture local and global features. To tackle this problem, the researchers have put forward a number of strategies, including those based on CNN and transformers. However, each of these solutions has its own limitations.

A. Semantic Segmentation of RS Images Based on CNN

Improving CNN's capacity to capture global dependence features has become a major focus of computer vision research in recent years. Existing methods primarily focus on expanding the receptive field of CNN, incorporating attention mechanisms, and integrating multiscale features [33], [34]. Various strategies have been proposed to broaden the receptive field of CNN, with the goal of improving their ability to capture and integrate larger contextual information from input data. For instance, atrous convolution, which introduces holes in the convolutional kernels to increase their receptive field, has been widely applied in the Deeplab series [35]. Pyramid pooling, used to capture global dependency features at different scales, as seen in PSPNet [36], aggregates features at various scales to effectively expand the receptive field. The introduction of attention mechanisms provides CNN with another means to selectively focus on important features. For example, SENet enhances channel feature recalibration through channel attention mechanisms, and nonlocal neural networks capture long-distance dependencies by computing responses across all positions in a feature map [37]. These attention mechanisms enhance the ability of the network to capture critical information and the accuracy of feature representation. However, these methods also have limitations: atrous convolution can lead to the gridding effect, where information in the feature map is unevenly distributed [38]. Pyramid pooling, while capturing multiscale information, may overlook finer details; attention mechanisms, though improving feature discriminability, can increase the computational burden and may be inefficient for large-sized inputs.

In the field of semantic segmentation of VHR RS images, CNN has made significant progress, particularly in enhancing the processing of global dependency features. Traditional CNN structures excel in handling local features but are limited in capturing global contextual information. Recent research has innovated various methods to bolster CNN in global dependency feature processing. Bertasius et al. proposed a neural field model, integrating FCN predictions with boundary clues to enhance semantic segmentation consistency and object localization accuracy [39]. BNF uses CNN convolutional filters for boundary detection and the predicted boundaries to define pairwise potentials in the energy model, thereby improving global segmentation performance. Nekrasov et al. [40] introduced global deconvolutional networks, obtaining dense predictions through global deconvolution operations to improve semantic segmentation models. This method incorporates global dependency features in locally extracted features, addressing CNN challenges in pixel-level label prediction. Moreover, Mou et al. [41] used spatial and channel relation modules to learn and infer global links, followed by relation-enhanced feature representations. These modules are versatile and can be combined with existing FCN frameworks in a plug-and-play manner. Finally, Wang et al. [42], [43] designed a dense upsampling convolution and hybrid dilated convolution framework to effectively expand the network's receptive field, and aggregate global dependency features. While existing methods have made progress in enhancing the ability of CNNS to capture global features, limitations persist, such as uneven expansion of the receptive field and neglect of global features [44].

B. Semantic Segmentation of RS Images Based on Transformers

In recent years, the advent of vision transformers (ViT) as an alternative to CNN has shown considerable potential in the field. By self-attention, the ViT model divides images into small patches, which are then entered sequentially into the

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model. Through its self-attention mechanism, the transformer model effectively captures long-range dependencies, offering a method to enhance global information extraction, an aspect traditionally challenging for CNN [45]. Moreover, ViT's architecture offers better scalability in processing large-sized images. While transformer-based models excel in capturing global dependency features, they face challenges in data and computational resource demands, local feature extraction efficiency, and optimization in handling features of varying scales. This is particularly pronounced when processing images with complex textures or details, where they lack an inductive bias for local neighborhoods. The Swin transformer, using a hierarchical structure and shifted windows, has set new benchmarks in various visual tasks, proving more efficient [46]. However, Swin transformer performance in dense-prediction segmentation tasks suboptimal, and a simplistic combination with existing methods could lead to increased size and parameters. To overcome these limitations, Shi et al. [47] proposed SSformer, a lightweight yet effective transformer model. Given the Swin transformer's inherent layered design, a decoder was introduced to combine input from various layers [47]. The transformer model, due to its unique attention mechanism, exhibits strong performance in tasks of global feature extraction within the context of semantic segmentation tasks. This capability improves the capacity to attend to and integrate information throughout the entire image, which helps it comprehend complicated scenarios more thoroughly.

In the semantic segmentation of VHR RS images, the application of ViT has also made significant progress. In recent years, advanced deep networks and AI models have significantly enhanced RS capabilities [48]. Notable examples include SpectralGPT, which utilizes transformer architectures for hyperspectral image analysis, and various advanced fusion models that integrate multiple data sources for improved segmentation accuracy. Wang et al. [49] introduced UNetFormer. A lightweight ResNet18 is used as the encoder in UNetFormer, and an effective global–local attention mechanism is added to the decoder. WiCoNet introduces an additional context branch to aggregate information from larger image areas, embedding context information into local features through a context transformer [50]. This approach overcomes the locality limitation of CNN, allowing WiCoNet to perceive a broader picture before segmenting land cover/land use (LCLU) categories. However, the precise combination of local features and global representations remains an unresolved issue. However, transformer models are still not as efficient as CNN in handling local details. For instance, the Laplacian-former method proposed by Azad et al. [51] aims to enhance the self-attention maps by adaptively recalibrating frequency information to overcome the limitations of ViTs in local texture detection. Although transformers excel in processing global information, they may fall short in capturing high-frequency components and local textures.

In summary, although CNN and transformers each have their strengths, they still face challenges when tasked with the semantic segmentation of RS images. For instance, CNN excels in capturing local details but may fall short in processing global information; on the other hand, transformers can handle global information but might have limitations in capturing local textures and edge details. Our approach seeks to combine the advantages of both CNN and transformers, proposing a dual-branch encoding structure that integrates transformers and CNN. The model's transformer branch gives users a macroscopic picture of all contextual data globally, which helps them understand the general arrangement and complex spatial interactions between scenes. Meanwhile, the CNN branch is dedicated to extracting local features, allowing the model to focus on specific terrain features and details. Overall, the aim is to enhance the precise extraction of global contextual information and edge features in VHR images.

III. Model Framework

A. Overall Architecture

In this article, a new dual-branch encoding network architecture—DESformer, as depicted in Fig. 3, is introduced. The main differences between DESformer and existing works are highlighted as follows. First, the dual-branch encoding structure consists of a CNN branch and a transformer branch, each tasked with capturing local features and global contextual information, respectively. This setup provides an efficient framework for multiscale feature fusion. The essence of the DESformer lies in the incorporation of an FIM, embedded within each encoding block to facilitate continuous semantic fusion between dual-branches. This dual-encoder configuration allows the network to simultaneously capture local and global dependency features and interactively reduce the semantic gaps between the two branches. This approach not only enhances the global perceptual capabilities of local features but also improves the local detail resolution of global representations. Furthermore, we introduce a centralized feature pyramid network module, MVI, for integrating local and global dependent features, effectively achieving a comprehensive and distinct feature representation. This innovative block structure further enhances the network's capability to process complex scenes in VHR images. Finally, we employ a composite loss function—FIAB-Loss, which integrates global and boundary loss functions. This composite loss function is designed to enhance the accuracy in segmenting small-scale objects and edge features while optimizing the ability to discriminate between different categories in land use the leveraging global dependency features.

Unlike existing dual-branch encoding networks, we introduce an FIM, which facilitates interactive fusion between local features and global representations at various resolutions (see Fig. 4). This structure not only enhances the capability to capture subtle features in RS images but also improves its understanding of large-scale spatial relationships. The central concept is to leverage the intrinsic multiscale, hierarchical structure of deep convolutional networks to build a feature pyramid that effectively captures features at various scales within an image. We incorporate the MVI module, based on FPN, as a crucial part of the model for integrating multilevel information. By constructing a multilevel feature pyramid, this module effectively merges features from various levels using a top-down approach as an upsampling process. This approach up-samples high-level feature maps, rich in semantic information, to higher resolutions and then merges them

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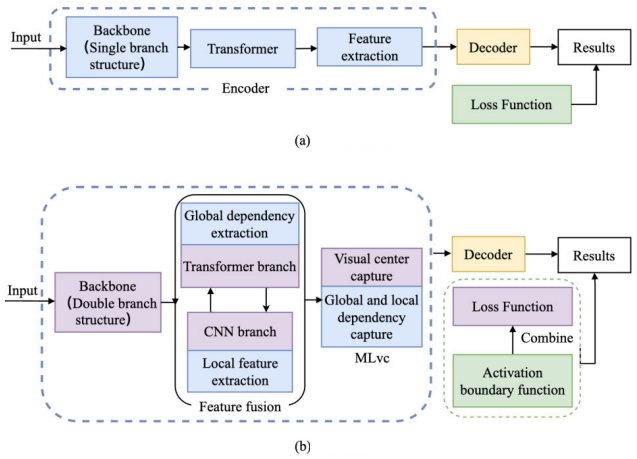


Fig. 3. Comparative work: in the existing methods, the single branch structure backbone network extracts global dependent features through a transformer, then extracts and decodes the features, and obtains the segmentation results through a single loss function. In contrast, our proposed DESformerr model adopts a two-branch coding structure. In order to achieve continuous semantic fusion between the two branches, we introduce an FIM module that integrates local features with global representations at varying resolutions. At the same time, the combined loss function for boundary detection is used to optimize the boundary. (a) Previous work. (b) Ours work.

with corresponding lower level feature maps. This structure ensures that each level's feature map contains rich semantic information while maintaining high spatial resolution. Such a multiscale feature fusion mechanism enables the model to handle both large-scale objects and small-scale details in images, thus enhancing the model's performance across different resolutions. The top layers of the FPN capture more abstract and global features, while the lower layers focus on more specific and local features.

Specifically, DESformer, the primary encoder structure consists of an initial layer and a dual-branch encoding network. A Conv2d layer with a 7×7 kernel, a stride of 2, and a padding of 3 is the main component of the first layer. This increases the number of input channels from 3 to 64. A BatchNorm2d layer with a ReLU activation function and a MaxPool2d layer with a 3×3 kernel and a stride of 2 are also included. Subsequent dual-branch encoding network blocks gradually increase channel numbers through a series of ConvBlocks, each containing a convolutional layer, batch normalization, and ReLU activation function, and incorporate residual connections to enhance training efficiency and network performance. The auxiliary encoder includes multiple ConvTransBlocks, each composed of a CNN block, fusion block, squeeze block, and expand block, supplemented by a trans block. The trans block includes standard components of a transformer: a multihead self-attention (MSA) mechanism, where QKV transformations map from 384 dimensions to 1152 dimensions, and a multilayer perceptron (MLP) with an intermediate layer of 1536 dimensions and an output layer of 384 dimensions. Each branch consists of N repeated CNN and transformer blocks. The FIM module employs an interactive approach to gradually integrate different types of features from the two branches; the MVI module effectively consolidates these features through a multiscale feature fusion mechanism. During training, we utilize a composite loss function to supervise the segmentation accuracy. This model not only benefits from both local features and global dependency features but also effectively integrates these through a multiscale feature fusion mechanism. This structure enhances the ability to recognize objects of different scales in RS images and also improves its understanding of complex scenes within these images. The combination of a dual-branch encoding structure with FPN provides a powerful and flexible solution for handling diverse visual tasks, particularly in applications that require simultaneous processing of features of varying scales and complex boundaries.

The primary challenges in the semantic segmentation of RS images include high intraclass variability and low interclass variability, complex, and irregular object boundaries, the need for effective multiscale feature integration, data imbalance, and the computational demands of processing large-scale data. DESformer addresses these challenges through a hybrid approach that combines CNN and transformers for enhanced feature extraction, the MLvc module for integrating multiscale features, and the composite FIAB-Loss function to handle class imbalance and improve boundary accuracy.

B. Dual-Branch Encoding Structure

1) CNN Branch: The convolutional branch focuses on extracting local feature details from the image, utilizing a standard ResNet encoder to capture local information. The CNN branch is broken into four phases, each with several convolutional blocks, as shown in the figure (CNN Branch). A 1×1 down-projection convolution, a 3×3 spatial convolution, a 1×1 up-projection convolution, and residual connections connecting the input and output of the bottlenecks are among the bottleneck structures established by ResNet that are included in each convolutional block [see Fig. 5(a)]. In the experimental setup, the first convolutional block is set as stage 1, while in the subsequent $N-1$ blocks, the number of bottlenecks is set to 2 or more. This design enables the CNN branch to effectively capture local features. In contrast to the ViT, which projects image patches directly onto vectors in a single step potentially losing local details, the convolutional kernels in the CNN branch slide over overlapping feature maps. This approach not only enhances feature locality but also allows for the preservation of detailed local features.

2) Transformer Branch: The transformer branch is important for extracting global features and follows the design principles of ViT. As shown in Fig. 3, the branch consists of N duplicate transformer blocks. During the tokenization phase, we use a linear projection layer to transform the feature map that the stem module produced into a 14×14 patch embedding. The standard transformer module [shown in Fig. 5(b)] consists of an MSA module and an MLP module. In each MSA module, an MSA allows the model to process information in parallel in different subspaces, thus capturing the global context. By breaking down input features into multiple heads, each of which calculates self-attention independently, and then aggregating the results, MSA modules are better able to learn complex global dependencies. The MLP module consists of fully connected (FC) layers that project up and down. These layers are processed by nonlinear activation functions, capable of capturing and representing complex patterns of

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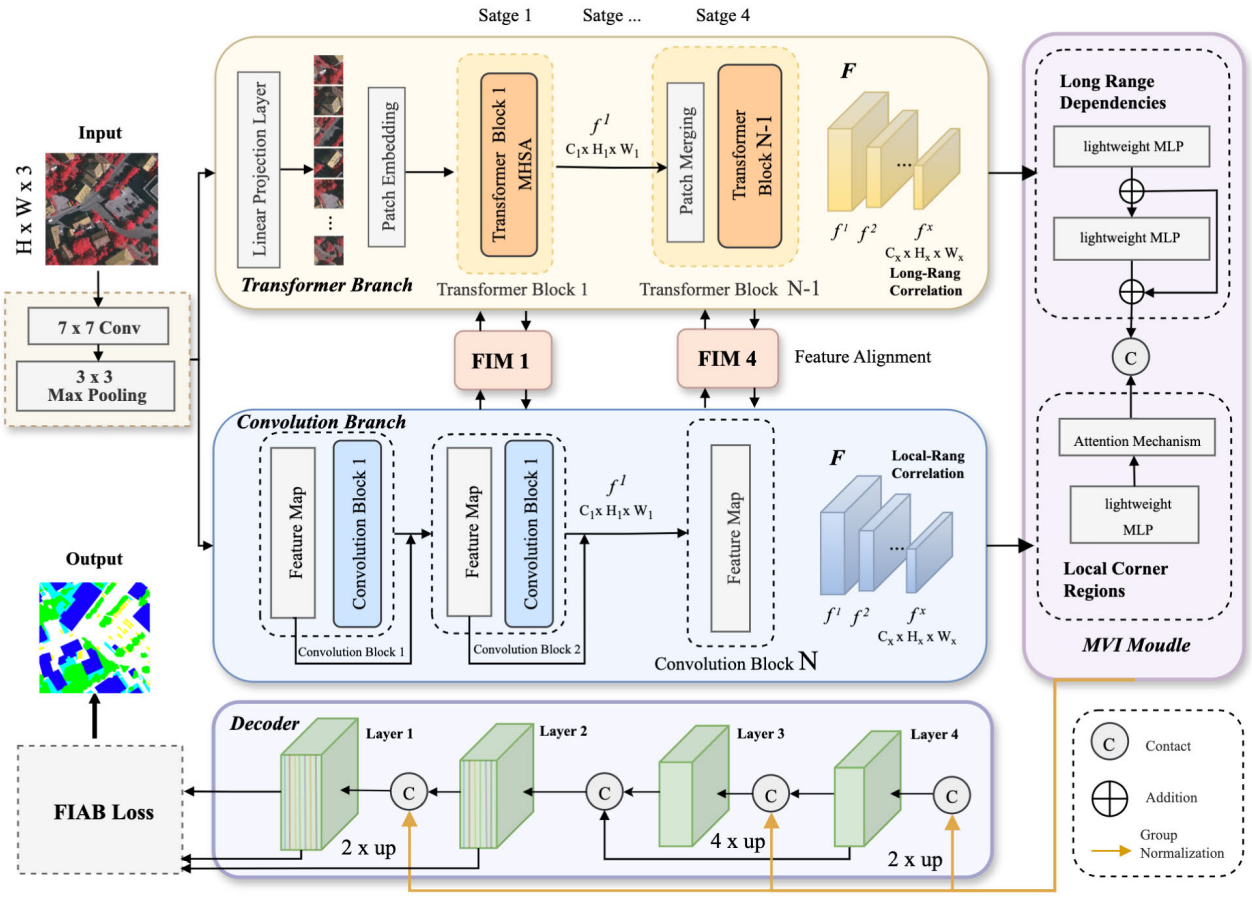


Fig. 4. Overview of dual-branch coding network architecture DESformer. It consists of the transformers branch, which extracts global features, and CNN, which extracts local features. Adding FIM modules to each coding block in the two branches solves the misalignment problem, and achieves continuous semantic fusion between the two branches. Next, the hierarchical feature pyramid with different resolutions is obtained through the dual-branch coded backbone network. The output is into the MVI module of the centralized feature pyramid network. The LCRs are composed of an attention mechanism and a lightweight MLP that capture global long-term dependencies. Long-range dependencies consist of two lightweight MLPS that aggregate information about local corners of input images. The entire MVI module enhances the ability to handle complex scenes in VHR images. Finally, multiscale feature fusion is carried out for the two branches. FIAB-Loss is a composite loss function, to improve the accuracy of small-scale objects and edge features.

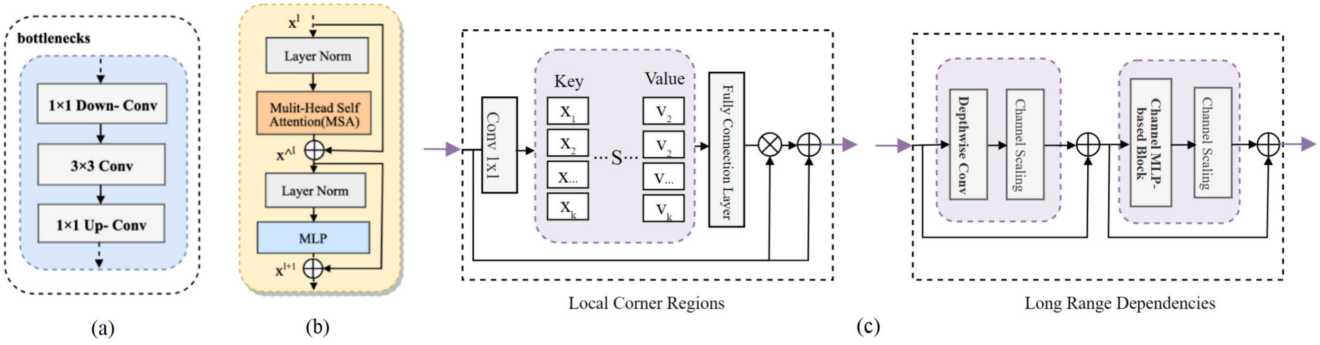


Fig. 5. (a) CNN branch module. (b) Transformer branch module. (c) MVI module. The MVI module aggregates local corner area information of the image using a lightweight MLP and a learnable vision center method.

input features. The upcast FC layer expands the dimension of the feature, making it more expressive, while the downcast FC layer restores it back to the original dimension, thus maintaining the consistency of the input and output dimensions. LayerNorm is used before each layer of the self-attention and MLP blocks and before the residual connection. The application of LayerNorm helps standardize the input at each layer, thus stabilizing the training process and reducing the problems of gradient disappearance and gradient explosion. In addition, LayerNorm enables the model to perform consistently across different batches of data, thereby improving the overall performance of the model. Through the transformer

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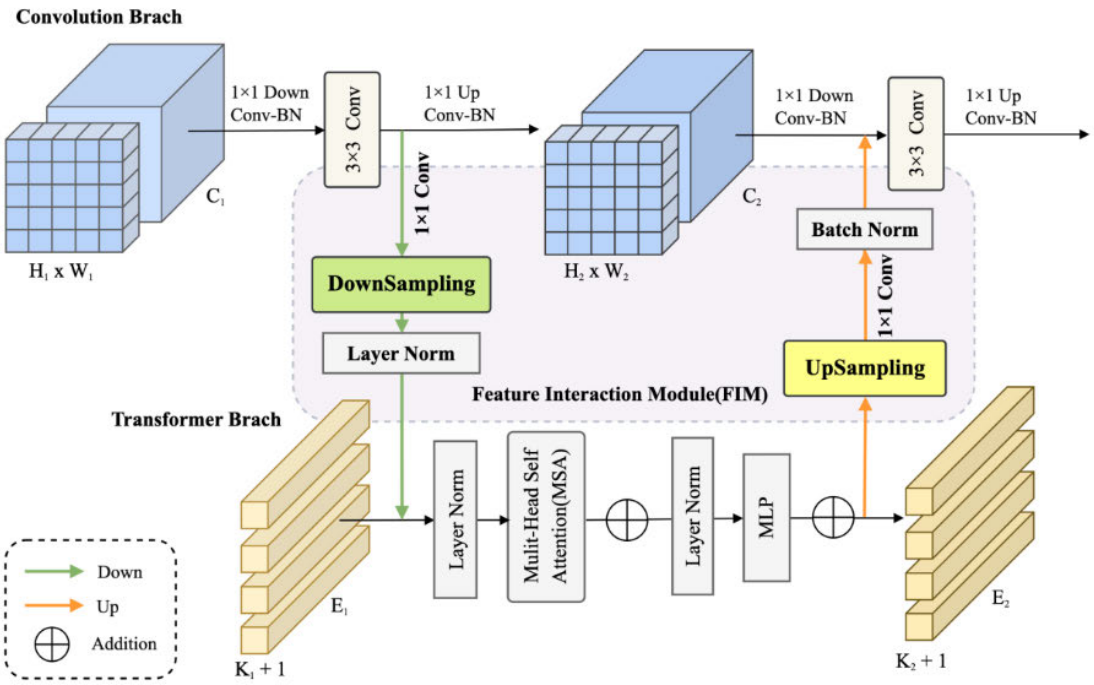


Fig. 6. FIM solves the alignment between the feature mapping of the CNN branch and the patch embedding of the transformer branch.

branch, it can effectively capture global and local information and perform well on complex visual tasks

$$\hat{x}^l = W_{\mathrm{MSA}}\left(\mathrm{LN}\left(x^{l-1}\right)\right) + x^{l-1} \tag{1}$$

$$x^l = \mathrm{MLP}\left(\mathrm{LN}\left(\hat{x}^l\right)\right) + x^l. \tag{2}$$

During the tokenization phase, we employ a nonoverlapping linear projection layer to transform the feature maps generated by the stem module into 14 × 14 patch embeddings. Subsequently, a class token, assumed to be identical to the patch embeddings, is utilized for segmentation tasks. Notably, since the CNN branch has already efficiently encoded local features and spatial position information through 3 × 3 convolutions, the transformer branch does not require additional positional embeddings. This design enhances the resolution processing capability of the image in downstream visual tasks. By employing an MSA, the block can process data in parallel across various representational subspaces, thereby enhancing feature expressiveness. It provides robust support for processing VHR images and complex visual tasks through efficient self-attention mechanisms and MLP blocks, working in tandem with the CNN branch. This method combines local feature extraction with global dependent feature processing through a double branch coding structure. This dual-branch design not only optimizes the efficiency of feature extraction but also enhances the understanding and processing ability of complex scenes.

3) Feature Interaction Module: The FIM is critical in resolving the alignment issue between the CNN feature maps and transformer patch embeddings. Initially, considering the inconsistency in feature dimensions, we implemented a series of measures to align them. The dimensions of CNN feature maps are $C \times H \times W$, where C stands for the number of channels, H for height, and W for width. In contrast, the shape of patch embeddings is $(K + 1) \times E$, with K representing the number of image patches, 1 denoting the class token, and E indicating the embedding dimension. To match the dimension of patch embeddings, we change the channel number of feature maps using 1 × 1 convolution (see Fig. 6). Subsequently, the subsampling module then aligns the spatial dimensions to convert the feature mapping into patch embedding. Next, the channel size embedded in the patch is adjusted by 1 × 1 convolution to match the number of channels in the CNN feature map. This convolution operation not only adjusts the number of channels but also optimizes the features, making them more suitable for merging with the CNN feature map. The adjusted patch embedding is integrated back into the CNN feature map. Moreover, we also employ LayerNorm and batch normalization modules to normalize the features. On the other hand, since CNN feature maps through local convolution operations and patch embeddings through global self-attention, there is a significant semantic difference between the two. To progressively bridge this semantic gap, FIM is applied to each block (except the first block).

The FIM plays a pivotal role in our network architecture, which bridges the semantic gap between local convolutional features and global self-attention features. The aligned features from the CNN and transformer are then fused through methods such as concatenation or element-wise addition, creating a unified feature representation that incorporates both local detail and global context. This fusion enhances the global perception capability of local features and retains the local details in global representations. The concurrent structure ensures that both local and global features are preserved, resulting in a model that generalizes better across different scales and types of objects, improving its robustness and performance in diverse RS scenarios. The primary objective of the FIM is to facilitate an interactive and continuous coupling

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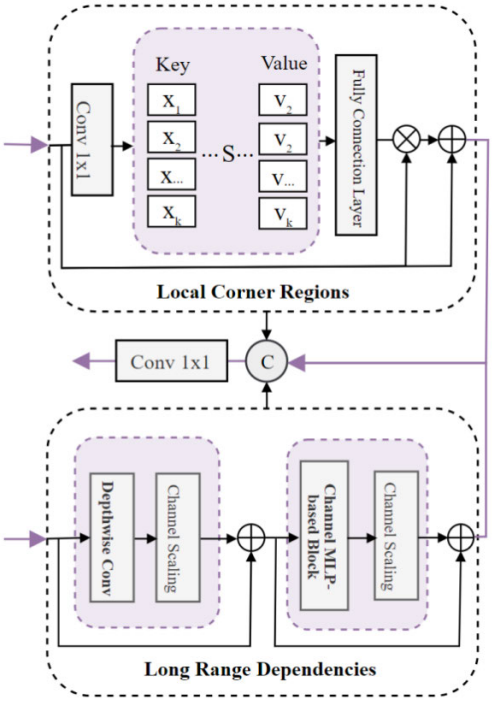


Fig. 7. Architecture of the MVI module. The MVI module consists of two main components: LCRs and long-range dependencies (LRD). LCRs: This component captures local features using a key-value mechanism and convolutional operations. The key-value pairs are processed through an FC layer to enhance the representation of LCRs. LRD: This component captures global dependencies using depthwise convolutions and channel scaling operations. It includes a channel MLP-based block to further refine feature representation. The outputs of both components are integrated through a concatenation operation and further processed to provide a comprehensive multiscale feature representation, improving the overall segmentation accuracy.

between local features and global representations, overcoming the dimensional and semantic discrepancies between the two types of features. By integrating these complementary strengths, DESformer achieves significant improvements in tasks like image classification, object detection, and instance segmentation, showcasing its potential as a general backbone network.

4) Visual Center Module: The implementation of the MVI module involves initially passing the input image through a dual-branch encoding backbone network to extract a hierarchical feature pyramid X (see Fig. 7). Based on this feature pyramid, we employed the concept of a concentrated feature pyramid (CFP). To effectively capture global long-range dependencies present at the top layer of the feature pyramid, X4, we proposed a lightweight MLP architecture. This simplified design reduces the consumption of computing resources, making it more efficient in practical applications. In addition, we employ a lightweight MLP in conjunction with a learnable visual center to aggregate data from the image's local corner areas. This learnable visual center mechanism dynamically adjusts and captures the most important feature areas in an image, thereby enhancing the model's focus on key parts. The network with the aforementioned parallel architecture, known as the MVI module, is designed to increase processing speed while ensuring highly accurate image parsing and feature extraction capabilities.

The MVI module employs a lightweight MLP to combine two parallel-connected blocks, specifically to capture global long-term dependencies in the top layer feature X4, i.e., global dependency features. This design allows the network to understand and process the broad spatial relationships in images. We introduced an innovative learnable visual center mechanism to precisely capture local details in the image. This mechanism effectively aggregates local corner area features on the X4 layer, thereby retaining important local features. The capture of such local features is crucial for understanding detailed features in images. We concatenate the feature maps that are produced along the channel dimension in order to build the MVI final output, which integrates the outputs of these two blocks. In our implementation, to facilitate a smoother feature transition between X4 and the MVI module, we used a Stem block. This stem block includes a 7×7 convolution layer, a batch normalization layer, and an activation function layer, ensuring smooth transitions and efficient feature integration.

LVC is an encoder featuring an inherent dictionary, as illustrated in Fig. 5(c). It consists of two primary components: the first part includes an inherent codebook, $B = \{b_1, b_2, \ldots, b_K\}$, where $N = H \times W$ represents the total spatial number of the input features, with H and W indicating the height and width of the feature map, respectively. The second part comprises a set of scaling factors, $S = \{s_1, s_2, \ldots, s_K\}$. They serve as hubs of learning about images. In particular, characteristics from the Stem block, X_{in}, are initially encoded through a combination of convolutional layers, including 1×1 convolution, 3×3 convolution, and another 1×1 convolution. After that, CBR blocks—which are made up of 3×3 convolutions coupled with BN layers and ReLU activation functions—process these encoded features. Following these steps, the encoded features, $\hat{X}_{\text{in}}$, are input into the codebook. To achieve this, we employ a series of scaling factors, s, sequentially mapping $\hat{x}_i$ and b_k to their respective spatial information. The information for the entire image in relation to the kth codeword is obtained as follows:

$$e_k = \sum_{i=1}^{N} \frac{e^{-s_k \|\hat{x}_i - b_k\|^2}}{\sum_{j=1}^{K} e^{-s_k \|\hat{x}_i - b_j\|^2}} \left(\hat{x}_i - b_k\right). \tag{3}$$

We used a top-down global concentration regulation (GCR) strategy to make sure the shallow features of the feature pyramid effectively utilize the concentrated visual information from the deepest layer features. Explicit visual center information acquired from the deepest layer features is used in this technique to make all previous shallower features (i.e., X3 to X0) adjustments. By integrating the lightweight MLP architecture with the GCR mechanism, the FPN implementation significantly enhances the feature pyramid's expressive power. This method not only improves the capability to recognize objects of various scales in images but also boosts overall network performance in complex visual tasks through precise feature fusion.

5) Boundary Optimization Combination Loss Function: In this research, we introduce a composite loss function—FIAB-loss combined with focal loss, IoU loss, and active boundary loss (ABL) to enhance the semantic segmentation performance of VHR RS images [52], [53]. This innovative loss function design aims to improve edge detection accuracy by focusing

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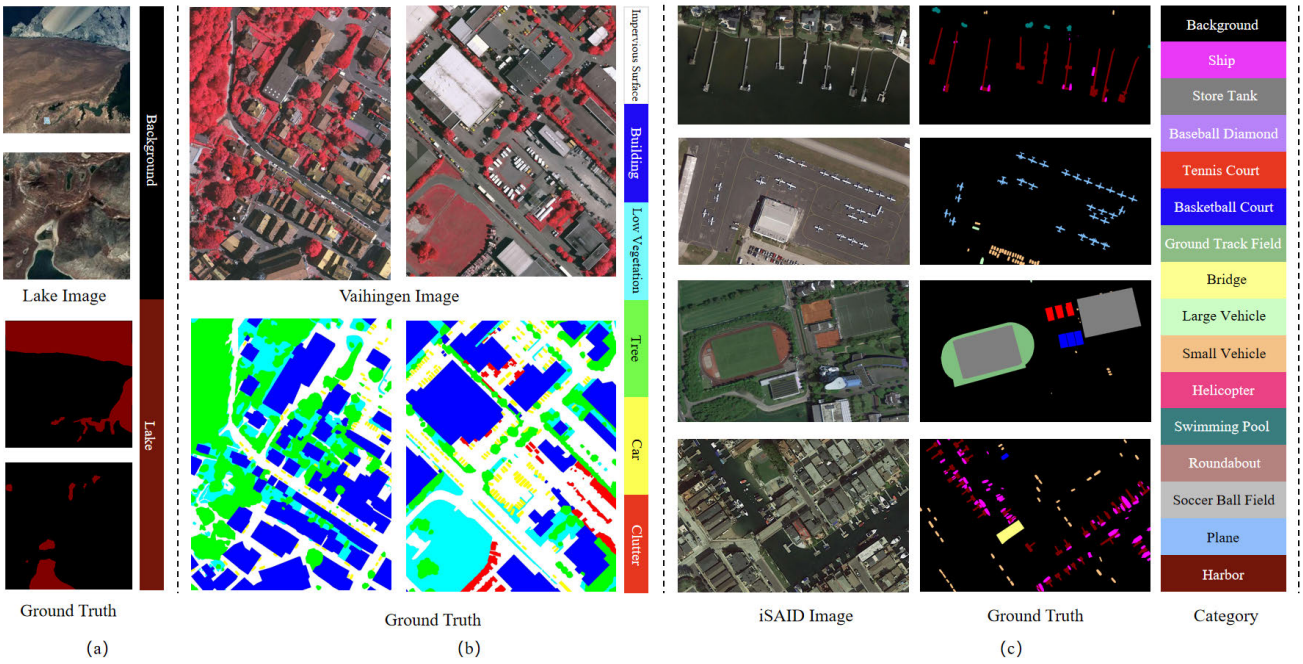


Fig. 8. Visual samples from different datasets. (a) Tibetan Plateau semantic segmentation lake dataset. (b) Vaihingen dataset. (c) iSAID dataset.

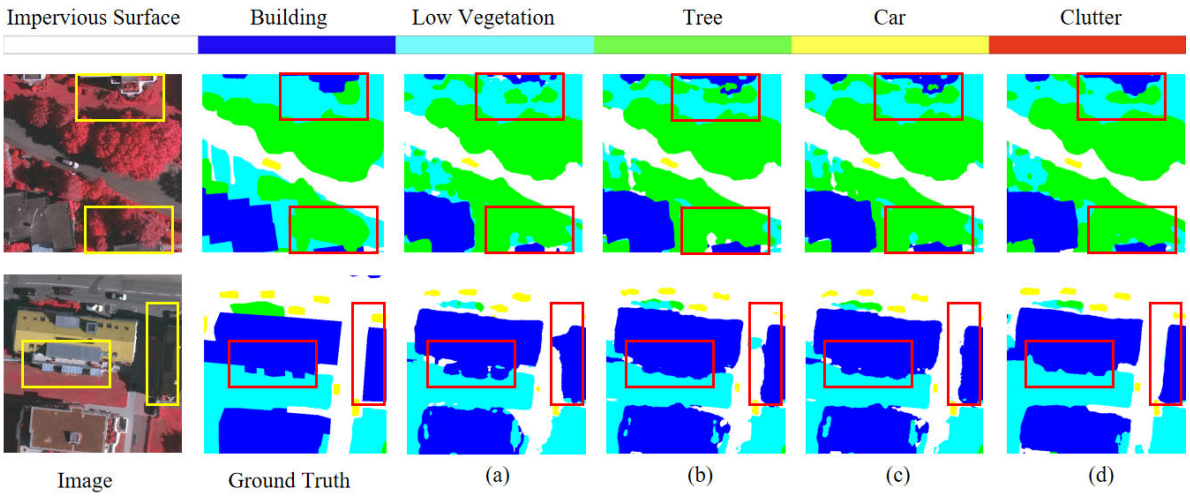


Fig. 9. Ablation experiments of dual-branch architecture on Vaihingen dataset. (a) Transformer. (b) CNN + Transformer (add). (c) CNN + Transformer (contact). (d) CNN + Transformer + FIM.

attention on hard-to-distinguish categories, thereby achieving more precise semantic segmentation. Focal loss primarily addresses the issue of class imbalance by emphasizing difficult and easily misclassified samples to enhance performance. The model is able to concentrate more on difficult samples by decreasing the contribution of easy-to-classify examples by altering the cross-entropy loss. IoU loss, commonly used in segmentation tasks, emphasizes the importance of the overlap area between predictions and actual annotations. IoU loss directly optimizes target boundaries, enhancing the precision of segmentation. ABL, specifically introduced to enhance the model's focus on predicted boundaries, is achieved by calculating the Kullback–Leibler (KL) divergence of neighboring pixels of predicted boundary pixels and generating a boundary map, thus identifying pixels that may belong to object boundaries. It also involves computing the nearest target direction from the predicted pixels to the ground truth boundaries and encoding this direction as a heat vector. By calculating the cross-entropy loss, the network is further encouraged to align with the actual ground truth boundaries

$$B_i = \begin{cases} 1, & \text{if } \mathrm{KL}(P_i, P_j) > e_j, \quad j \in \mathcal{N}_2(i) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$D_i^g = \Phi\left(\arg\min_j M_{i+1+j}\right), \quad j \in \{0, 1, \ldots, 7\} \tag{5}$$

$$D_i^p = \frac{e^{\mathrm{KL}(P_i, P_{i+A_k})}}{\sum_{m=0}^{7} e^{\mathrm{KL}(P_i, P_{i+A_m})}}, \quad k \in \{0, 1, \ldots, 7\} \tag{6}$$

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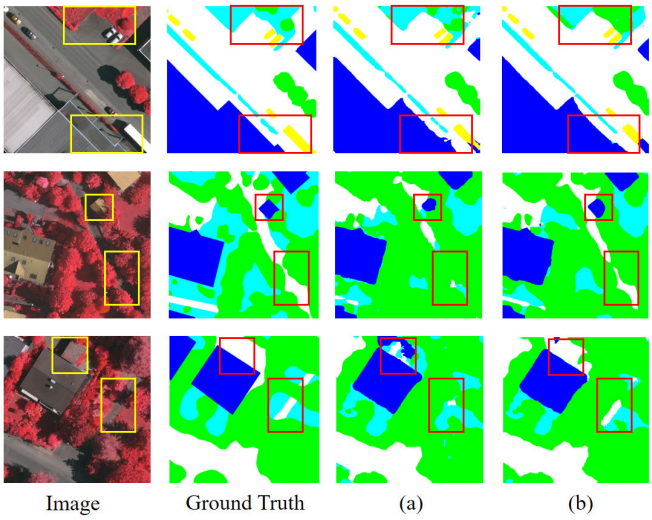


Fig. 10. Ablation experiment of LRD part of MVI module on Vaihingen dataset. (a) CNN + Transformer + FIM + LRD. (b) CNN + Transformer + FIM + MVI.

$$\mathrm{ABL} = \frac{1}{N_b}\sum_i A(M_i)\mathrm{CE}\left(D_i^p, D_i^g\right). \tag{7}$$

By integrating these three types of loss functions, our composite loss function is designed to enhance the performance in VHR RS image semantic segmentation through various mechanisms. Focal loss intensifies the focus on difficult-to-classify samples, IoU loss directly optimizes segmentation boundaries, and ABL further emphasizes accurate prediction of boundary pixels. This multidimensional loss function combination provides an effective approach to addressing the complex task of semantic segmentation in VHR RS images

$$\mathrm{IoU} = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}}\Delta_c(m(c)) \tag{8}$$

$$\mathrm{FL}(p_t) = -(1-p_t)\log(p_t) \tag{9}$$

$$L_t = \alpha\mathrm{FL} + \beta\mathrm{IoU} + \theta\mathrm{ABL}. \tag{10}$$

IV. Experiments and Results

A. Datasets

1) Tibetan Plateau Lake Semantic Segmentation Dataset: The Tibetan Plateau lake dataset is a semantic segmentation dataset focused on lake water bodies across the Tibetan Plateau, comprising RGB images that cover the entire region (see Fig. 8) [54], [55], [56]. The dataset contains a total of 6774 images with a size of 256 × 256. The dataset is labeled with two categories. In the experiment, we used 20% of the data, or 1355 images, for testing, and the remaining 80%, or 5419 images, for training. In this study, we focus on the edge detection of lake water bodies. Lake boundaries show irregular shapes and changing profiles, which puts forward higher requirements for edge detection accuracy and can better verify the accuracy and robustness of the model when dealing with complex natural landscapes.

2) Vaihingen Dataset: The Vaihingen dataset includes 33 RS images of varying sizes [57]. The selection process ensured the avoidance of data voids. The top orthophoto and DSM have a spatial resolution of 9 cm. This dataset, annotated with six categories, is utilized for semantic segmentation studies, featuring three channel combinations per image: IR-RG, RGB, and RGB-IR. For the experiment, 20% of the images were used for testing, the remaining 80% were used for training, and all the original images were adjusted to 512 × 512 pixels.

3) iSAID Dataset: The iSAID dataset is designed for instance segmentation of aerial images [58], [59]. This dataset contains 15 categories with a total of 655 451 object instances spread across 2806 VHR images. Notable features of iSAID include: a) a vast collection of high spatial resolution images; b) 15 significant and common categories; c) numerous instances per category; d) extensive labeling of instance images facilitating contextual learning; e) significant variation in object scale within the same image; f) imbalanced and uneven distribution of objects with differing orientations that reflect real-world aerial conditions; and g) numerous small objects with ambiguous appearances that require contextual reasoning for resolution. For the experiment, all the original images were adjusted to 896 × 896 pixels.

B. Experimental Details

1) Detailed Experimental Setup: In this research, we conducted experiments on scene semantic segmentation using the Mmsegmentation, which is based on the PyTorch framework. The hardware used included the NVIDIA Tesla V100S PCIe with 32 GB of memory. The experimental setup is detailed as follows: we selected stochastic gradient descent (SGD) as the optimizer, setting the initial learning rate to 0.01, the momentum to 0.9, and the weight decay factor to 0.0005. These parameter settings accelerate convergence during the initial stages of training and use weight decay to manage model complexity, thereby preventing overfitting. The learning rate scheduling strategy adopted was polynomial decay (PolyLR), which gradually reduces the learning rate from its initial value to 0.0001 according to (11), where power = 0.9, thereby adjusting the learning rate throughout the training process

$$\mathrm{lr} = \mathrm{lr}_{\mathrm{initial}} \times \left(1 - \frac{\mathrm{iter}}{\mathrm{max_iter}}\right)^{\mathrm{power}}. \tag{11}$$

The entire training period was set for 80 000 iterations to ensure sufficient depth of training and enhance the generalization capability. An iterative training loop (IterBasedTrainLoop) was employed with a maximum of 80 000 iterations, and validation (ValLoop) was conducted every 8000 iterations to monitor and evaluate the performance of an independent dataset. In addition, the experiment incorporated various hooks functionalities, including the IterTimerHook for monitoring the duration of each iteration, LoggerHook for logging information every 50 iterations, ParamSchedulerHook for managing the learning rate schedule, and CheckpointHook for saving the model every 8000 iterations. This comprehensive experimental setup aims to finely control multiple key parameters during the training process, ensuring the model achieves optimal performance under strict experimental monitoring. Periodic validation and regular model saving effectively assess and document the model's performance at various stages.

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TABLE I
ABLATION EXPERIMENTS OF DUAL-BRANCH ENCODING NETWORK ARCHITECTURE ON THE VAIHINGEN DATASET

Method	Connection Type			IoU (%) per Category					Metrics	
	Series	Parallel	FIM	Imp.surf	Buildings	Lowveg	Trees	Cars	mIoU	mF1
CNN	-	-	-	77.71	85.59	64.87	75.98	66.32	74.09	83.74
Transformer	-	-	-	85.49	90.24	65.86	73.05	60.69	75.06	84.76
CNN+Transformer (add)	-	✓	-	82.49	89.24	65.86	75.05	66.69	75.86	84.52
CNN+Transformer (contact)	✓	-	-	85.86	88.63	64.74	77.01	67.61	76.77	88.17
CNN+Transformer+FIM	-	-	✓	**87.05**	**87.27**	**67.46**	**76.05**	**75.85**	**78.73**	**88.32**

2) Evaluation Metrics: We use the average $F1$ score and the mean intersection over union (mIoU) to assess model performance. The confusion matrix, which has four components—true positives (TPs), false positives (FPs), true negatives (TNs), and false negatives (FNs)—is the source of these two measurements. A TP occurs when a pixel or area is correctly classified as belonging to a category. An FP occurs when a model incorrectly classifies a pixel as belonging to a certain category. A TN represents pixels that are accurately identified as not being part of a specific category, whereas an FN denotes pixels that are incorrectly classified as not belonging to that category. The following is the computation method:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{12}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

$$\text{mIoU} = \frac{1}{Z} \sum_{n=1}^{Z} \frac{\text{TP}_n}{\text{TP}_n + \text{FP}_n + \text{FN}_n} \tag{14}$$

$$\text{mF1} = \frac{2}{Z} \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{15}$$

where precision is $(\text{TP}/(\text{TP} + \text{FP}))$ and recall is $(\text{TP}/(\text{TP} + \text{FN}))$. mIoU represents the average of IoU across all classes. The mF1 score is the average of $F1$ scores across all categories.

C. Ablation Experiment

We conducted a series of extensive ablation tests on the Vaihingen dataset in order to evaluate the performance of the DESformer network design and its fundamental elements.

1) Dual-Branch Encoding Network Architecture: In order to evaluate the effectiveness of various network topologies and talk about the experimental performance of single-branch and dual-branch models, we undertook a number of ablation experiments in this section. As shown in Table I, we compared the effects of a single-branch model with a dual-branch model that combines CNN and transformer. To further demonstrate the effectiveness of the FIM, we conducted additional ablation experiments isolating its impact (see Fig. 9). These experiments compared the performance of the model with and without the FIM module while keeping the two-branch network constant. The results, presented in Table I, indicate that the FIM module leads to a measurable improvement in segmentation accuracy by enhancing the interaction and fusion of local and global features.

The results indicate that the CNN model, serving as a baseline, performed particularly well in segmenting buildings and trees, achieving intersection over union (IoU) scores of 85.59% and 75.98%, respectively. However, after combining CNN with a transformer in either a serial [labeled as CNN + Transformer (add)] or parallel [labeled as CNN + Transformer (concat)] configuration, there was a significant improvement in performance across multiple categories. Specifically, in the serial configuration, although the IoU for buildings slightly decreased, there was an increase in performance for vehicles and the overall mean IoU (mIoU), which reached 75.86%, an improvement of 1.82% over using CNN alone. This suggests that the transformer added valuable contextual information that helped improve the overall prediction accuracy. In the parallel configuration, the model showed improvements in almost all categories, particularly notable in the vehicle and tree categories, with IoU scores increasing to 77.01% and 67.61%, respectively. The mIoU and mean $F1$ (mF1) scores in this configuration reached 76.77% and 88.17%, indicating that the parallel structure more effectively integrates features from both CNN and transformer, optimizing feature representation and enhancing the prediction accuracy and robustness. These experimental results demonstrate the effectiveness of using a two-branch structure to design semantic segmentation networks for RS images. This structure not only enhances the understanding of complex spatial relationships but also improves recognition capabilities for small targets such as vehicles.

2) MVI Attention Extraction Module: In Table II, we present the ablation experiments of the MVI module on the Vaihingen dataset. To provide a clear benchmark for comparison, the first column lists the results of the "CNN + Transformer + FIM" configuration. Subsequent experiments will illustrate the incremental performance gains achieved by integrating the LRD and LCR components of the MVI module. The final "CNN + Transformer + FIM + MVI" configuration demonstrates the most significant improvements across various metrics, validating the effectiveness of the MVI integration.

As shown in Table II, the results of the ablation experiments reveal the contributions of LRD and LCR both individually and in combination to network performance (see Figs. 10 and 11). In the configuration where LRD was removed (CNN + Transformer + FIM + LCR), the model showed a slight decline in overall performance, with mIoU and mF1 scores of 75.67% and 89.62%, respectively. This indicates that LRD is crucial for capturing long-range dependencies, which helps to enhance the understanding of long-distance interactive

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TABLE II
ABLATION EXPERIMENTS OF THE MVI MODULE ON THE VAIHINGEN DATASET

Method	MVI Block		IoU (%) per Category					Metrics	
	LRD	LCR	Imp.surf	Buildings	Lowveg	Trees	Cars	mIoU	mF1
CNN	-	-	77.71	85.59	64.87	75.98	66.32	74.09	83.74
CNN+Transformer+FIM	-	-	87.05	87.27	67.46	76.05	75.85	78.73	88.32
CNN+Transformer+FIM+LRD	✓	-	82.91	92.13	70.86	78.35	77.67	80.38	89.89
CNN+Transformer+FIM+LCR	-	✓	83.91	93.13	72.86	75.35	75.67	80.18	89.62
CNN+Transformer+FIM+MVI	✓	✓	**86.91**	**92.13**	**70.86**	**79.35**	**78.04**	**81.45**	**90.12**

TABLE III
PROPOSED FIM MODULE STAGE JUMP FUSION ABLATION EXPERIMENT ON THE VAIHINGEN DATASET

Method	Stage1	Stage2	Stage3	mIoU (%)	mF1 (%)
DESformer-FIM1	-	-	-	79.97	87.28
DESformer-FIM2	✓	-	-	80.16	88.69
DESformer-FIM3	✓	✓	-	80.95	89.55
DESformer-FIM4	✓	✓	✓	**81.45**	**90.12**

features in images. In contrast, when using only LRD and removing LCR (CNN + Transformer + FIM + LRD), the performance improved in some respects, with mIoU and mF1 scores of 77.67% and 89.89%, respectively. This demonstrates that LRD plays a more significant role in handling local feature corner region information, effectively enhancing the ability to capture details. When both components are used simultaneously (CNN + Transformer + FIM + MVI), the model performs optimally, achieving a mIoU of 78.04% and an mF1 of 90.12%. This result emphasizes the importance of the combined action of LRD and LCR, which together can significantly enhance the ability to process both global and local information, thereby providing more accurate segmentation results in complex scenes.

3) FIM Module Stage Jump Fusion: In this study, we conducted a series of ablation experiments to explore the effects of the FIM applied at different stages within the dual-branch encoding structure, aiming to assess its role in extracting global contextual information and improving overall model performance. The experimental design involved systematically increasing the number of FIMs from the first to the third stage to investigate the specific contributions of FIMs at each stage.

While multiscale features have been extensively explored in many works [49], [60], [61], [62], [63], [64], our approach introduces the MLvc module, which uniquely integrates multiscale features from different levels of the network. The results in Table III align with the well-established benefits of multiscale features but specifically highlight the improvements brought by the MVI module in our DESformer framework. This module enhances segmentation accuracy by capturing both fine-grained local details and broader contextual information, demonstrating the effective integration of multiscale features in our model.

As shown in Table III, the DESformer-FIM0, serving as the baseline model without any FIMs, achieved mIoU and mF1 scores of 79.97% and 87.28%, respectively. With the introduction of FIM in the first stage (DESformer-FIM1), we observed an improvement in model performance, with mIoU and mF1 increasing to 80.16% and 88.69%, respectively. This indicates that introducing FIM at an early stage effectively facilitates the capture of more refined features, thereby enhancing its ability to understand complex scenes. Furthermore, with FIM introduced at both the first and second stages (DESformer-FIM2), the model's mIoU and mF1 were again elevated, reaching 80.95% and 89.55%, respectively. This result demonstrates that FIM plays a positive role in consecutive stages, enhancing feature interaction between stages and improving the model's integration of global and local information. Finally, when FIM is applied at all three stages (DESformer-FIM3), the model exhibits its best performance, with mIoU and mF1 reaching 81.45% and 90.12%, respectively. This significant improvement emphasizes the importance of applying FIM across multiple stages for synthesizing and optimizing global contextual information, particularly when dealing with complex spatial relationships in images.

These experimental results validate the crucial role of introducing FIM at various stages in enhancing model segmentation accuracy and handling complex image features. Visually, models with FIM more effectively distinguish between object occlusions and similarities, thereby reducing misidentifications and improving segmentation accuracy. Furthermore, these findings provide valuable insights for future image processing model design, particularly in how FIMs can be utilized to optimize model performance. In Fig. 12, we have used images of different sizes (1024 × 1024) to demonstrate the robustness and effectiveness of the DESformer models. By including these variations, we show that the models can effectively handle various input scales, which is crucial for real-world applications where RS images often come in different sizes. The consistent performance across different image dimensions further emphasizes the adaptability and reliability of our models, ensuring that they maintain high segmentation accuracy regardless of the image scale.

These experimental results validate the crucial role of introducing FIM at various stages in enhancing model segmentation accuracy and handling complex image features. Visually, models with FIM more effectively distinguish between object occlusions and similarities, thereby reducing misidentifications and improving segmentation accuracy. Furthermore, these findings provide valuable insights for future image processing model design, particularly in how FIMs can be utilized to optimize model performance.

4) Loss Function FIAB-Loss Module: In this section, the Vaihingen dataset was used for a series of ablation experiments to test and examine the effects of various combined loss

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TABLE IV
PERFORMANCE COMPARISON UNDER DIFFERENT DATA CONDITIONS

Method	Imp.surf	Buildings	Lowveg	Trees	Cars	mIoU	mF1
White Gaussian Noise	85.28	90.84	70.45	80.26	76.84	80.73	89.06
Conventional Dataset	89.68	92.31	72.84	79.69	78.47	82.59	90.92

TABLE V
SEMANTIC SEGMENTATION RESULTS ON THE VAIHINGEN DATASET

Method	Backbone	Imp.surf		Buildings		Lowveg		Trees		Cars		mF1	mIoU
		F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	(%)	(%)
FCNs[65]	VGG-16	90.94	79.22	93.71	83.97	62.18	50.87	73.19	60.38	42.67	38.92	72.53	62.67
DeeplabV3[66]	ResNet-50	89.40	81.25	89.05	86.56	76.60	55.43	89.53	71.86	56.39	62.24	80.19	71.46
DeeplabV3plus[67]	ResNet-50	86.03	84.97	80.68	90.01	64.61	66.28	89.56	75.80	66.98	75.37	87.57	78.48
PSPNet[68]	ResNet-50	89.77	83.24	93.93	89.22	79.79	69.54	89.85	73.32	79.32	69.81	86.53	77.02
Segformer[69]	MIT-B0	91.20	82.49	94.41	89.35	78.33	67.66	89.21	78.50	88.54	73.30	88.33	78.26
CCNet[70]	ResNet-50	90.08	79.07	90.65	87.74	74.93	55.37	85.70	70.55	78.16	71.79	83.90	72.90
DANet[71]	ResNet-50	90.28	79.88	95.22	87.96	76.46	59.18	87.96	73.21	80.63	75.68	86.11	75.18
Semantic FPN[72]	ResNet-50	90.52	80.76	94.31	89.72	74.85	61.61	84.38	68.27	78.18	72.57	84.44	74.58
EncNET[18]	ResNet-50	88.19	79.64	90.85	88.13	74.09	60.54	83.24	68.32	78.54	70.57	82.98	73.44
MaskFormer[73]	ResNet-50	92.51	85.39	96.94	**93.32**	79.89	65.59	89.91	77.94	84.21	78.59	88.69	80.16
Mask2Former[74]	ResNet-50	95.07	89.46	92.58	90.83	81.92	71.97	88.35	75.51	**89.95**	76.78	89.57	80.91
UNetformer[49]	ResNet-18	92.33	85.76	96.25	92.78	80.47	67.33	90.85	83.22	89.35	80.75	89.85	81.97
CMTFNet[63]	ResNet-50	92.53	86.09	**96.95**	94.09	79.98	66.64	90.22	82.19	89.87	**81.60**	89.91	82.12
DESformer	Conformer	**95.29**	**89.68**	96.19	92.31	**82.55**	**72.84**	**91.44**	**79.69**	89.14	78.47	**90.92**	**82.59**

TABLE VI
SEMANTIC SEGMENTATION RESULTS ON THE ISAID DATASET

Method	Backbone	IOU per category (%)															mIoU	mF1
		Ship	ST	BD	TC	BC	GTF	Bridge	LV	SV	HC	SP	RA	SBF	Plane	Harbor	(%)	(%)
FCNs[65]	VGG-16	40.73	52.43	20.81	71.43	30.60	28.43	8.50	8.43	40.60	11.32	30.60	48.60	58.73	65.81	40.60	37.18	46.63
DeeplabV3[66]	ResNet-50	60.71	52.78	70.10	84.14	53.98	59.44	30.64	59.64	39.15	31.65	40.86	65.39	71.70	76.86	49.35	56.40	65.85
DeeplabV3+[67]	ResNet-50	62.03	68.83	73.57	85.83	57.57	**65.83**	34.52	61.33	33.57	30.83	45.35	67.57	72.87	81.57	50.57	59.49	69.43
Segformer[69]	MIT-B0	63.70	65.68	75.55	86.58	61.66	60.70	35.84	60.06	48.50	38.70	42.97	**69.51**	71.85	82.58	54.59	61.23	70.55
PSPNet[68]	ResNet-50	65.93	56.90	75.21	87.09	58.20	60.92	32.56	59.77	43.20	36.90	45.25	68.21	72.19	83.20	54.13	59.98	69.95
CCNet[70]	ResNet-50	60.45	65.46	68.90	82.90	57.10	56.80	34.00	57.60	38.30	31.60	36.59	57.20	75.10	75.80	45.90	56.23	67.27
DANet[71]	ResNet-50	61.87	63.12	71.40	84.70	50.90	52.58	28.60	57.95	42.14	30.40	46.61	40.66	63.30	80.63	48.80	54.91	63.16
Semantic FPN[72]	ResNet-50	63.58	59.50	71.87	86.55	57.89	51.60	34.21	59.20	45.10	10.05	46.89	68.17	73.60	80.07	52.42	57.38	66.75
EncNET[18]	ResNet-50	61.96	65.96	72.33	84.22	56.23	50.65	39.81	58.29	43.98	34.84	42.40	65.80	72.98	82.17	55.86	59.17	68.57
MaskFormer[73]	ResNet-50	64.48	74.41	70.20	84.78	57.08	53.41	40.05	59.41	49.52	43.41	40.20	63.28	73.41	85.60	52.20	60.76	70.20
Mask2Former[74]	ResNet-50	65.97	73.99	71.20	81.05	58.20	56.90	40.18	59.90	48.10	42.90	43.20	63.27	72.90	86.28	55.82	61.32	71.50
DESformer	Conformer	**65.97**	**75.15**	**75.65**	**87.15**	**62.20**	60.19	**40.20**	**61.65**	**49.99**	**42.90**	**47.56**	63.27	**76.19**	**87.28**	**56.20**	**63.43**	**72.98**

* Ship (ST), Store Tank (ST), Baseball Diamond (BD), Tennis Court (TC), Basketball Court (BC), Ground Track Field (GTF), Bridge, Large Vehicle (LV), Small Vehicle (SV), Helicopter (HC), Swimming Pool (SP), Roundabout (RA), Soccer Ball Field (SBF), Plane, Harbor

functions on model performance. The experimental design included the separate and combined use of IoU loss, focal loss, and ABL, aimed at exploring the effectiveness of these loss functions in improving model segmentation performance.

The results show that when using IoU loss alone, the model achieved a mIoU of 88.07%. When IoU loss was combined with focal loss (IoU + Focal), the mIoU increased to 89.16%, indicating that the introduction of focal loss helps to improve the model's learning efficiency for difficult-to-classify pixels, thereby enhancing overall segmentation accuracy. The experimental configuration incorporating ABL (IoU + ABL) exhibited a mIoU of 88.95%, suggesting that ABL has a positive effect on enhancing the model's prediction accuracy in boundary areas, although its improvement effect was not as significant as that of Focal loss. Finally, when IoU loss, focal loss, and ABL were used simultaneously (IoU + Focal + ABL), the model's performance peaked, with a mIoU increasing to 90.12%. This result highlights the importance of combining loss functions, which significantly enhances the model's overall performance by simultaneously

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TABLE VII
SEMANTIC SEGMENTATION RESULTS ON THE TIBETAN PLATEAU LAKE DATASET

Method	Backbone	Lake		Background		mF1 (%)	mIoU (%)
		F1(%)	IoU(%)	F1(%)	IoU(%)		
FCNs[65]	VGG-16	90.98	78.16	90.50	76.19	90.74	77.18
DeeplabV3[66]	ResNet-50	89.50	88.16	92.02	86.19	90.76	87.18
DeeplabV3plus[67]	ResNet-50	96.50	93.39	93.04	94.94	94.77	94.17
Segformer[69]	MIT-B0	92.20	91.86	95.62	93.12	93.92	92.49
PSPNet[68]	ResNet-50	91.21	91.35	93.04	88.80	92.13	90.08
CCNet[70]	ResNet-50	91.28	90.37	93.32	90.39	92.30	90.38
DANet[71]	ResNet-50	96.32	88.08	91.21	89.55	93.77	88.82
Semantic FPN[72]	ResNet-50	90.98	90.66	89.87	91.24	90.43	90.95
EncNET[18]	ResNet-50	94.81	91.87	91.05	92.12	92.93	92.00
MaskFormer[73]	ResNet-50	95.33	92.21	95.05	94.80	95.19	93.51
Mask2Former[74]	ResNet-50	96.92	93.08	94.68	94.55	95.80	93.82
DESformer	Conformer	**99.06**	**94.94**	**97.30**	**98.05**	**98.57**	**96.50**

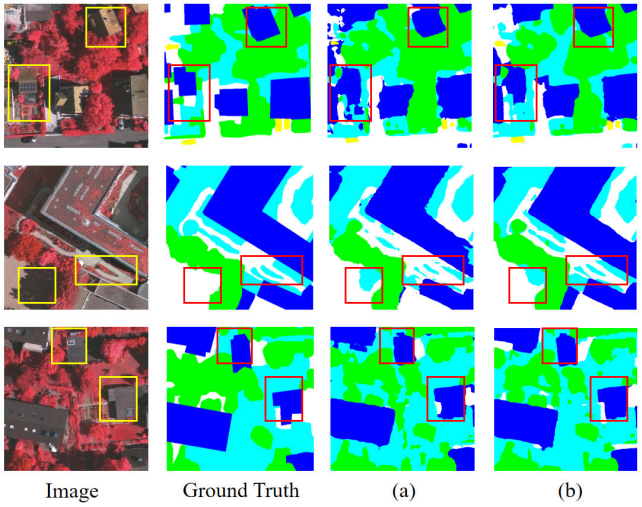


Fig. 11. Ablation experiment of LCR part of MVI module on Vaihingen dataset. (a) CNN + Transformer + FIM + LCR. (b) CNN + Transformer + FIM + MVI.

optimizing for different issues such as class imbalance and unclear boundary definitions.

Fig. 14 presents the ablation experiment results for the FIAB-Loss function, illustrating the impact of each component—focal loss, IoU loss, and ABL—on the segmentation performance of the DESformer model. The results indicate that Fig. 12(b) ensures accurate overlap between predicted and ground truth boundaries, providing a strong baseline for boundary accuracy. Fig. 12(c) enhances the model's precision in capturing complex boundaries and fine details, leading to improved delineation of intricate shapes. Fig. 12(d) configuration demonstrates the highest performance, effectively addressing class imbalance, boundary precision, and fine detail capture. This is evident from the consistently accurate segmentation in the red-highlighted areas. The results validate the importance of each loss component and their synergistic effect in improving overall segmentation accuracy, thereby demonstrating the effectiveness of the FIAB-Loss function in complex RS image segmentation tasks (see Fig. 13). These ablation experiment results emphasize the effectiveness of using a combination of multiple loss functions in complex RS image segmentation tasks. Focal loss improves overall segmentation outcomes by increasing the model's attention to minority classes, while ABL further enhances accuracy by refining the prediction in boundary areas.

5) Robustness and Adaptability: To simulate real-world imaging conditions where data can be affected by various degradation factors, we introduced Gaussian white noise with a standard deviation of 5 to the dataset. This setup aimed to evaluate the resilience of our DESformer model against common noise effects encountered during data acquisition (Table IV).

As shown in the attached table, the addition of white Gaussian noise resulted in a slight performance degradation compared to the conventional dataset. The mIoU decreased from 82.59% to 80.73%, and the mean $F1$ score (mF1) reduced from 90.92% to 89.06%. Despite these reductions, the model maintained high performance across all categories, with minimal drops in individual class IoU scores. For instance, the IoU for buildings decreased from 92.31% to 90.84%, while the IoU for cars decreased from 78.47% to 76.84%.

These results demonstrate that DESformer exhibits strong robustness to noise, maintaining competitive accuracy even under challenging conditions. The model's ability to handle noise can be attributed to its advanced MLvc integration, which effectively captures global and local features, thereby enhancing its resilience to spectral variability and noise.

D. Comparison With Other Methods

1) Results on the Vaihingen Dataset: Our DESformer model demonstrates superior performance across several metrics on the ISPRS Vaihingen dataset (see Fig. 15). Utilizing the

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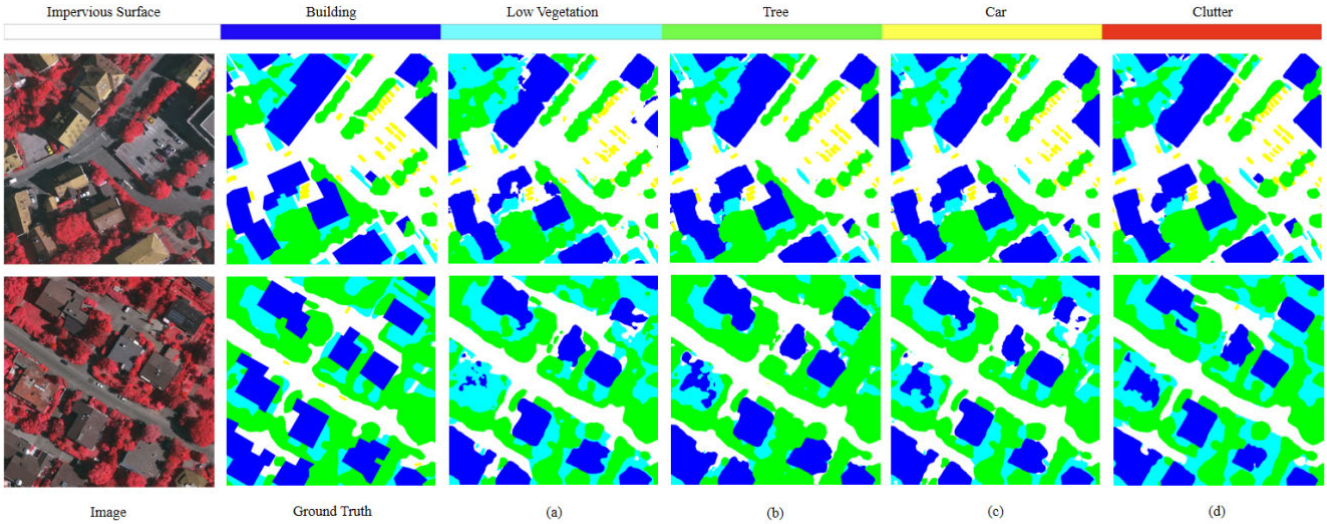


Fig. 12. Ablation experiment of FIM module stage jump fusion on the Vaihingen dataset. In this experiment, images of size 1024 × 1024 were used to demonstrate the robustness and effectiveness of the DESformer models. (a) DESformer-FIM0. (b) DESformer-FIM1. (c) DESformer-FIM2. (d) DESformer-FIM3.

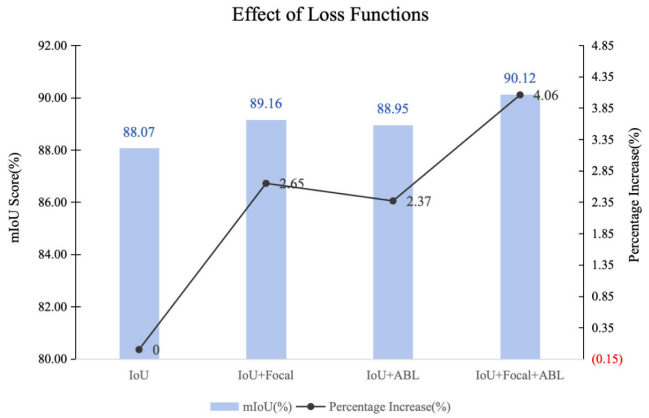


Fig. 13. Ablation experiment of loss function FIAB-Loss module on Vaihingen dataset.

TABLE VIII
EFFICIENCY ANALYSIS BETWEEN DIFFERENT MODELS

Method	Input Shape	FLOPs(G)	Params(M)
FCNs[65]	3x512x512	203.21	47.21
DeeplabV3[66]		126.98	35.42
DeeplabV3+[67]		181.20	41.22
PSPNet[68]		183.37	46.60
CCNet[70]		206.11	47.53
DANet[71]		217.58	47.68
EncNET[18]		144.25	33.66
MaskFormer[73]		270.54	44.32
Mask2Former[74]		265.98	43.44
DESFormer		69.71	54.40

transformer branch, the DESformer effectively captures global dependencies, enhancing the segmentation of complex scenes like low vegetation and trees. For low vegetation, DESformer achieved an $F1$ score of 82.55% and an IoU of 72.84%, while for trees, it reached an $F1$ score of 91.44% and an IoU of 79.69% (Table V). These scores surpass those of the second-best model, Mask2Former, by 3.56% and 3.87%, respectively. In building classification, DESformer attained an $F1$ score of 96.19% and an IoU of 92.31%, outperforming DeeplabV3+ by 2.3%. The model's dual-branch structure, which integrates local and global features, excels in preserving structural details and capturing complex urban environments. This architecture also proves beneficial in road detection, where DESformer achieved an $F1$ score of 95.29% and an IoU of 89.68%, ensuring accurate segmentation even in intricate urban layouts. In vehicle detection, DESformer reached an $F1$ score of 89.14% and an IoU of 78.47%, outperforming Mask2Former in both road and vehicle categories by 3.74% and 2.09%, respectively. In addition, DESformer surpassed recent SOTA models in multiclass segmentation tasks. Specifically, it outperformed UNetFormer in building segmentation and CMTFNet in low vegetation. It also excelled in tree classification, outperforming RS3 Mamba with an IoU of 79.69% and an $F1$ value of 91.44%, compared to RS3 Mamba's 67.43% IoU and 80.84% $F1$. Overall, using this double-branch structure, DESformer not only improved in each individual index but also performed exceptionally well in the comprehensive evaluation metrics. Its mF1 reached 90.92% and its mIoU value reached 82.59%, significantly surpassing other models. These comprehensive metrics indicate the model's robustness and effectiveness in various segmentation tasks.

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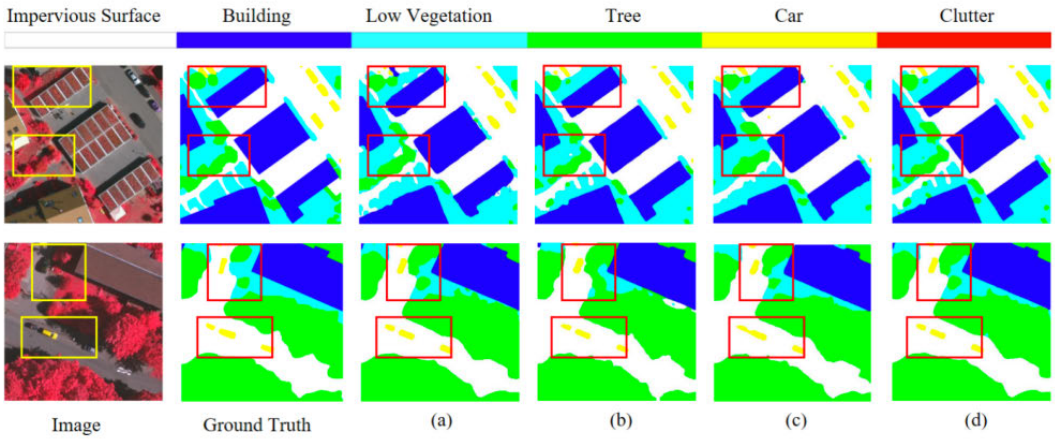


Fig. 14. Ablation experiment of FIAB-Loss function module on the Vaihingen dataset. (a) Focal loss. (b) IoU loss. (c) ABL. (d) FIAB-Loss.

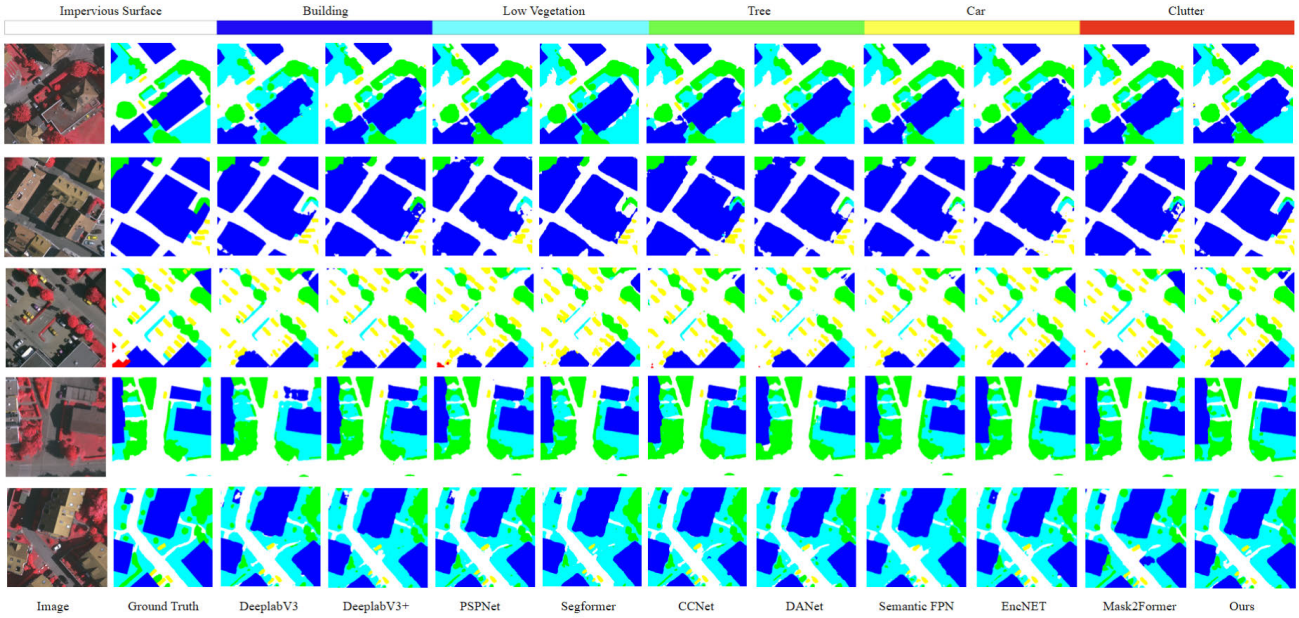


Fig. 15. Examples of semantic segmentation results on the Vaihingen dataset.

2) Results on the iSAID Dataset: According to the experimental results on the iSAID dataset, our model DESformer demonstrates notable improvements in small-size target recognition (see Fig. 16). DESformer effectively integrates global and local information through the MVI module, performing exceptionally well across multiple categories (Table VI). For example, in the detection of small-size targets such as bridges and small vehicles (SV), DESformer's IoU values reached 40.20% and 61.65%, respectively, compared with 34.01% and 59.41% for Mask2Former, representing increases of 6.19% and 2.24%. This improvement is attributed to the MVI module's capability to better capture global dependencies and detailed features, providing the model an edge in handling complex backgrounds and small targets. When examining the category of bridges, DESformer's enhanced performance is particularly beneficial due to the typically small and narrow nature of bridge structures within the dataset. The model's ability to maintain structural integrity and accurately delineate these features highlights its robustness in detecting elongated and thin objects. However, in densely built urban areas, minor bridge components may still be occasionally misclassified, indicating a potential area for further enhancement. For SVs, DESformer's IoU of 61.65% showcases its proficiency in distinguishing between small, similarly sized objects within cluttered environments. The slight improvement over Mask2Former underscores the model's precision in identifying and segmenting individual vehicles, even when they are closely packed. Nonetheless, challenges remain in differentiating vehicles from other small, spectrally similar objects, such as dumpsters or small trailers. DESformer also excels at handling complex boundaries and small targets, thanks in part to the design of its combined boundary loss function. When detecting categories such as baseball diamonds (BDs) and swimming pools (SPs), DESformer's IoU values were 75.65% and 47.56%, respectively, compared with 70.36% and 43.41% for CCNet, representing increases of 5.29%

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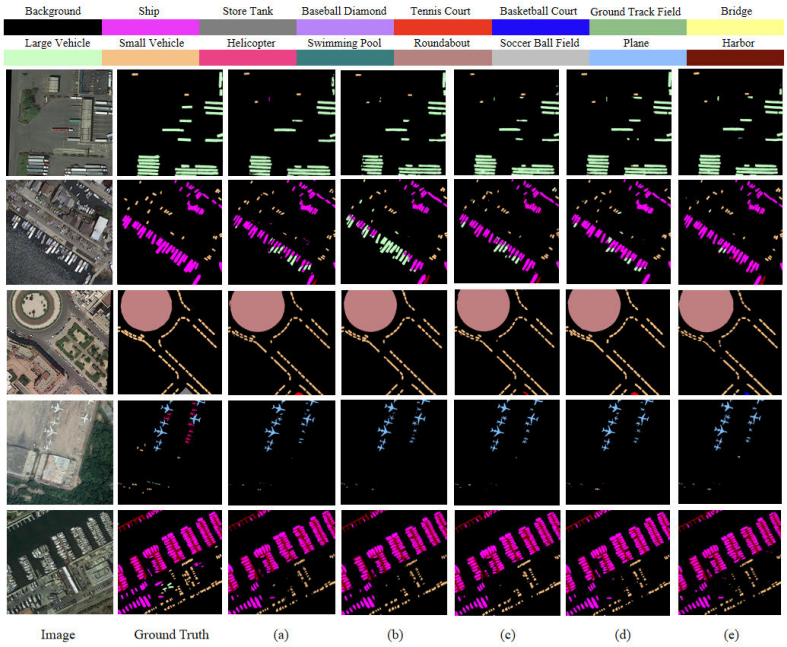


Fig. 16. Examples of semantic segmentation results on the iSAID dataset. (a) PSPNet. (b) Segformer. (c) DeeplabV3+. (d) Mask2Former. (e) DESformer.

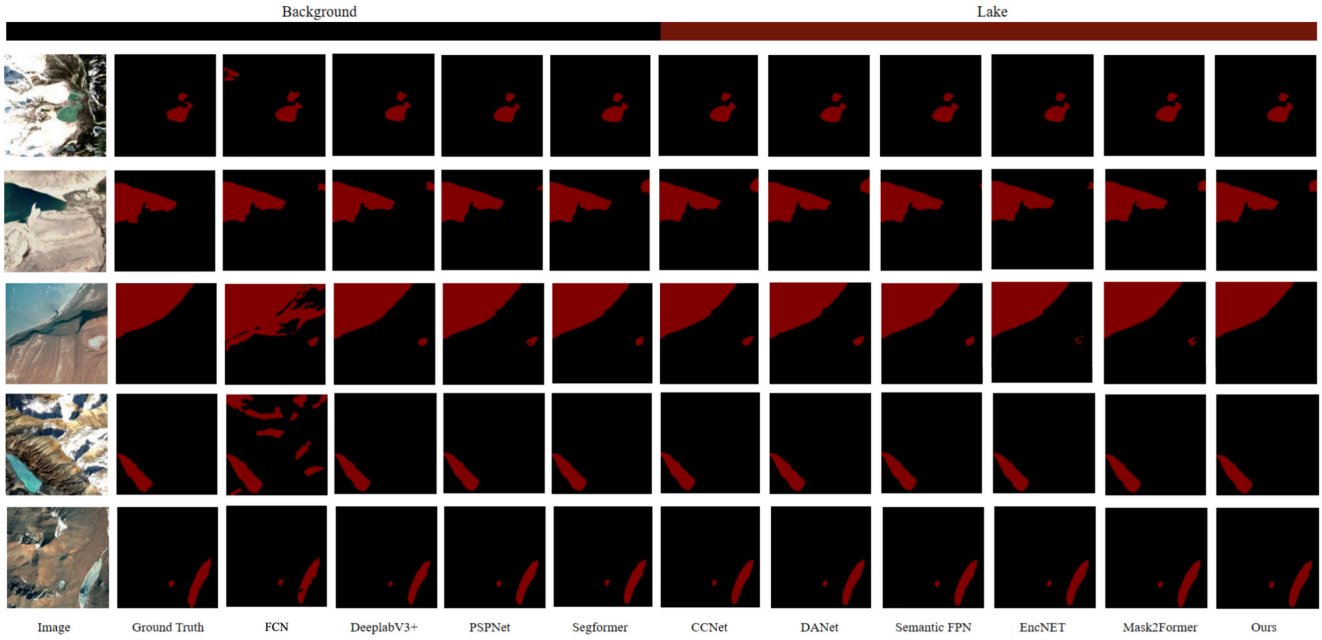


Fig. 17. Examples of semantic segmentation results on the Tibetan Plateau lake dataset.

and 4.15%. This notable increase in performance demonstrates how our boundary loss function successfully enhances the model's accuracy while handling small targets and complicated borders. For BDs, the model's ability to precisely capture the distinctive diamond shape and surrounding features, such as bases and outfield boundaries, is crucial. DESformer's performance in this category indicates its effectiveness in recognizing and segmenting complex geometric shapes within diverse backgrounds. However, variability in the appearance of BDs, due to differences in field maintenance and surrounding infrastructure, can still pose challenges. In the case of SPs, the model benefits from the clear spectral signature of water, which aids in accurate identification and segmentation. The improvement over CCNet highlights the DESformer's superior ability to handle intricate boundary details, such as pool edges and adjacent objects. Nevertheless, variations in pool shapes and sizes, as well as occlusions from surrounding structures, can occasionally lead to segmentation inaccuracies. Overall, DESformer achieves significant performance improvements in small-size object detection and complex boundary processing by combining the MVI module with the boundary loss function. The value of mIoU is 63.43%, and the value of

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mF1 is 72.98%, which exceeds other models in many indexes, verifying its superiority in RS image segmentation tasks. These enhancements are particularly evident in categories where the unique characteristics of the dataset, such as object size variability and complex boundaries, play a significant role.

3) Results on the Tibetan Plateau Lake Semantic Segmentation Dataset: According to the experimental results on the Tibetan Plateau lake dataset, our model DESformer demonstrates excellent performance in boundary detection processing (see Fig. 17). This advantage is mainly due to the boundary loss function incorporated in its design. From the result data, DESformer's IoU value for lakes reaches 94.94%, while the $F1$ value is as high as 99.06% (Table VII). In contrast, the second-placed Mask2Former has an IoU value of 93.32% and an $F1$ value of 96.92% in the lake category, showing improvements of 1.62% and 2.14%, respectively. This improvement indicates that the DESformer can process complex boundary information more accurately and carefully when extracting and delineating lake boundaries. In the lake category, DESformer's enhanced performance is particularly beneficial due to the complex and irregular shapes of lake boundaries within the dataset. The model's ability to maintain boundary integrity and accurately capture the fine details of these shapes highlights its robustness in detecting intricate and variable lake perimeters. In the performance of the background class, DESformer also shows excellent ability. This significant performance improvement can be attributed to the DESformer's use of a boundary loss function, which enhances the model's accuracy in dealing with background and lake boundaries, effectively avoiding fuzzy and inaccurate boundary segmentation. DESformer's precise boundary delineation ensures that background areas are accurately segmented, even when adjacent to complex lake edges. Overall, the mIoU value and mF1 value of DESformer on the Tibetan Plateau lake dataset reached 96.50% and 98.57%, respectively, significantly surpassing other models. This result demonstrates the effectiveness of the boundary loss function in increasing boundary processing accuracy, particularly when handling intricate fine boundaries.

E. Efficiency Analysis

To conduct a thorough assessment of our enhanced DESformer model, the table presents a comparison of the computational complexity and parameter count with other sophisticated models operating under the same conditions (Table VIII). In particular, we perform a thorough comparison of the number of parameters for each model and utilize floating-point computation (FLOPs) to quantify the computational complexity of the model, a decrease in which corresponds to a faster computing speed. The results show that compared with the traditional CNN-based model, our improved DESformer not only has obvious advantages in the number of parameters and computational complexity but also significantly improves the performance. We note that although DESformer has a higher number of parameters and a relatively long inference time, this is mainly due to the expansion of the MVI module. After introducing more parameters and increasing computational costs, our DESformer achieves a significant performance improvement compared to the original single-branch network.

V. Conclusion

In this research, we introduce a dual-branch coding structure DESformer, which combines CNN and transformer to improve the accuracy of VHR RS images. The core innovation of DESformer lies in its FIM, which effectively combines the advantages of the CNN branch in capturing fine local features with the powerful capability of the transformer branch in handling global dependencies. In addition, the MVI block embedded in the model provides global centralized supervision, significantly enhancing the recognition ability of different scales and small objects. After experimental validation on several widely used RS image datasets, DESformer outperforms existing advanced methods in performance. Our proposed FIM, MVI block, and combinatorial loss function are shown to be effective through in-depth ablation tests and contrastive analysis. The remarkable improvement in the segmentation accuracy fully shows the high efficiency and practicability of this method.

Acknowledgment

The Vaihingen dataset was provided by the German Society for Photogrammetry, Remote Sensing and Geoinformation (DGPF) [Cramer, 2010]: http://www.ifp.unistuttgart.de/dgpf/DKEP-Allg.html. Degradation pattern and coupling mechanism of mountain, water, forest, farmland, lake, grass and sand system in southern hilly region. Competent authority: Ministry of Science and Technology, People's Republic of China.

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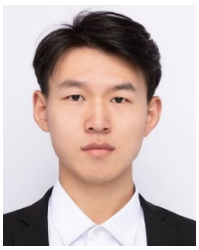
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